

FeedLact Experiment: Methanol Feeding Strategy Optimization

β -Lactoglobulin Production in Komagataella phaffii

EXPERIMENT OVERVIEW

Objective: Optimize methanol feeding strategy for recombinant protein production

Design: 2 strains $\times$ 3 strategies = 6 runs

- Strains: KP-B1 (baseline), KP-B2 (optimized)
- Strategies: Continuous, Pulsed, Ramped
- Duration: 120 hours (5 days)
- Replicates: 1 per condition

Product: Recombinant β -lactoglobulin

Process: Two-phase fed-batch

- Phase 1 (0-48h): Glucose growth
- Phase 2 (48-120h): Methanol induction

KEY RESULTS

Optimal Combination:

- ✓ Strain: KP-B2
- ✓ Strategy: Ramped
- ✓ Final titer: 33.2 g/L
- ✓ Improvement: 15.2% vs worst

Performance Range:

- Protein: 28.8 - 33.2 g/L
- Biomass: 18.5 - 24.0 g/L
- Viability: 90.4 - 93.4%

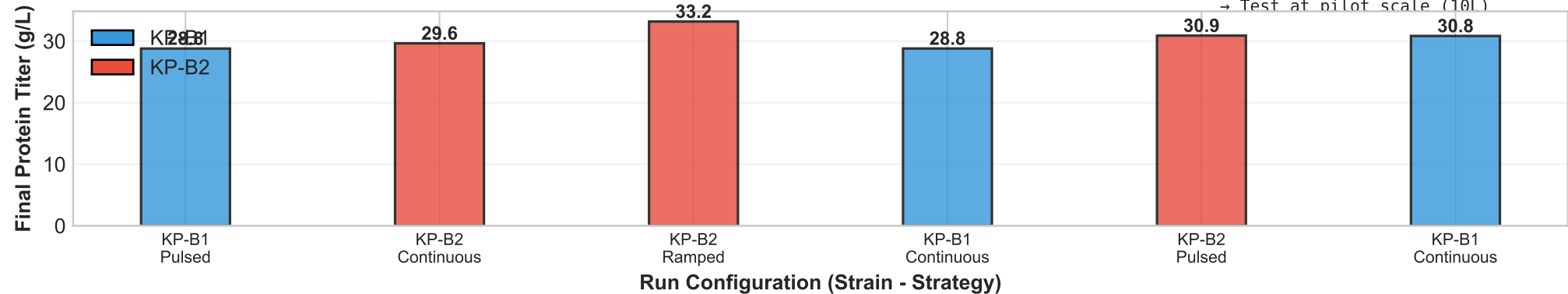
Statistical Significance:

- Strain effect: SIGNIFICANT
- Strategy effect: SIGNIFICANT
- Interaction: MODERATE

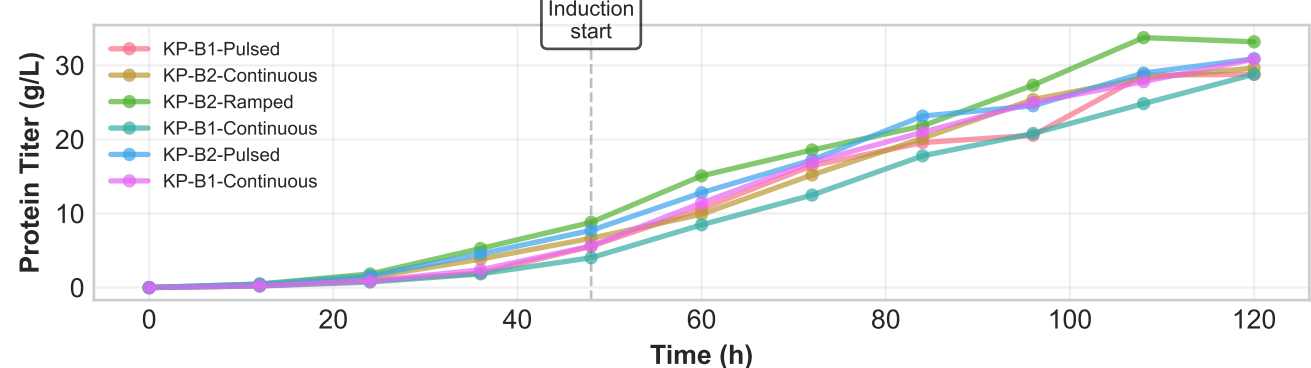
RECOMMENDATIONS

- SCALE-UP PARAMETERS
 - Use: KP-B2 strain
 - Strategy: Ramped feeding
 - Expected titer: ~33.2 g/L
- PROCESS OPTIMIZATION
 - Fine-tune Ramped parameters
 - Monitor viability (maintain >95%)
 - Optimize induction timing
- FURTHER STUDIES
 - Test intermediate feeding rates
 - Evaluate DO control strategies
 - Optimize temperature profile
- RISK MITIGATION
 - Validate with replicates (n $\geq$ 3)
 - Test at pilot scale (10L)

Final Protein Production Across All Conditions



Protein Production Time Course



Summary Statistics

Strain	Strategy	Protein (mg/L)	Biomass (g/L)
B1	Cont	29821	23
B1	Puls	28793	24
B2	Cont	29639	18
B2	Puls	30895	19
B2	Ramp	33180	19